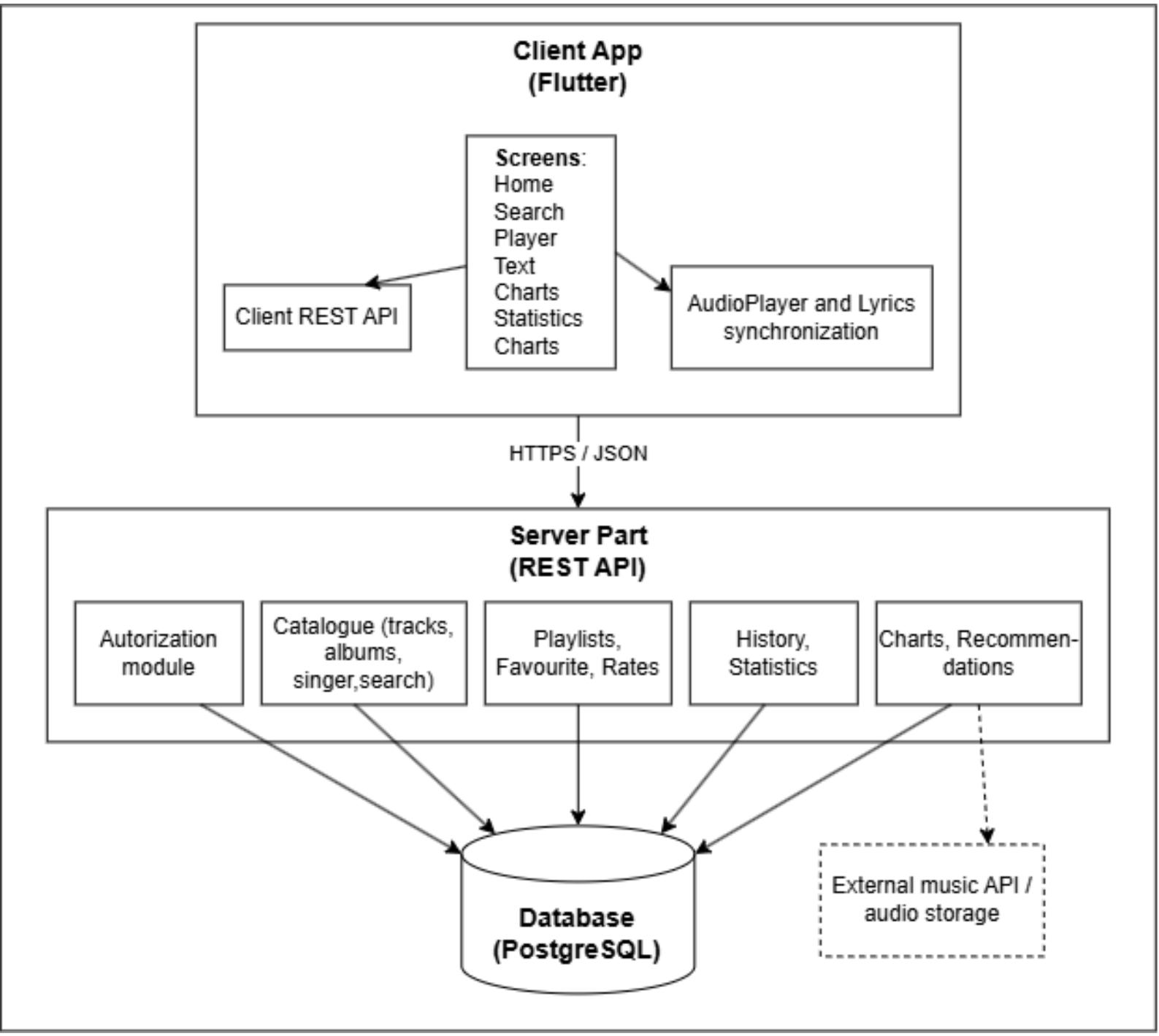
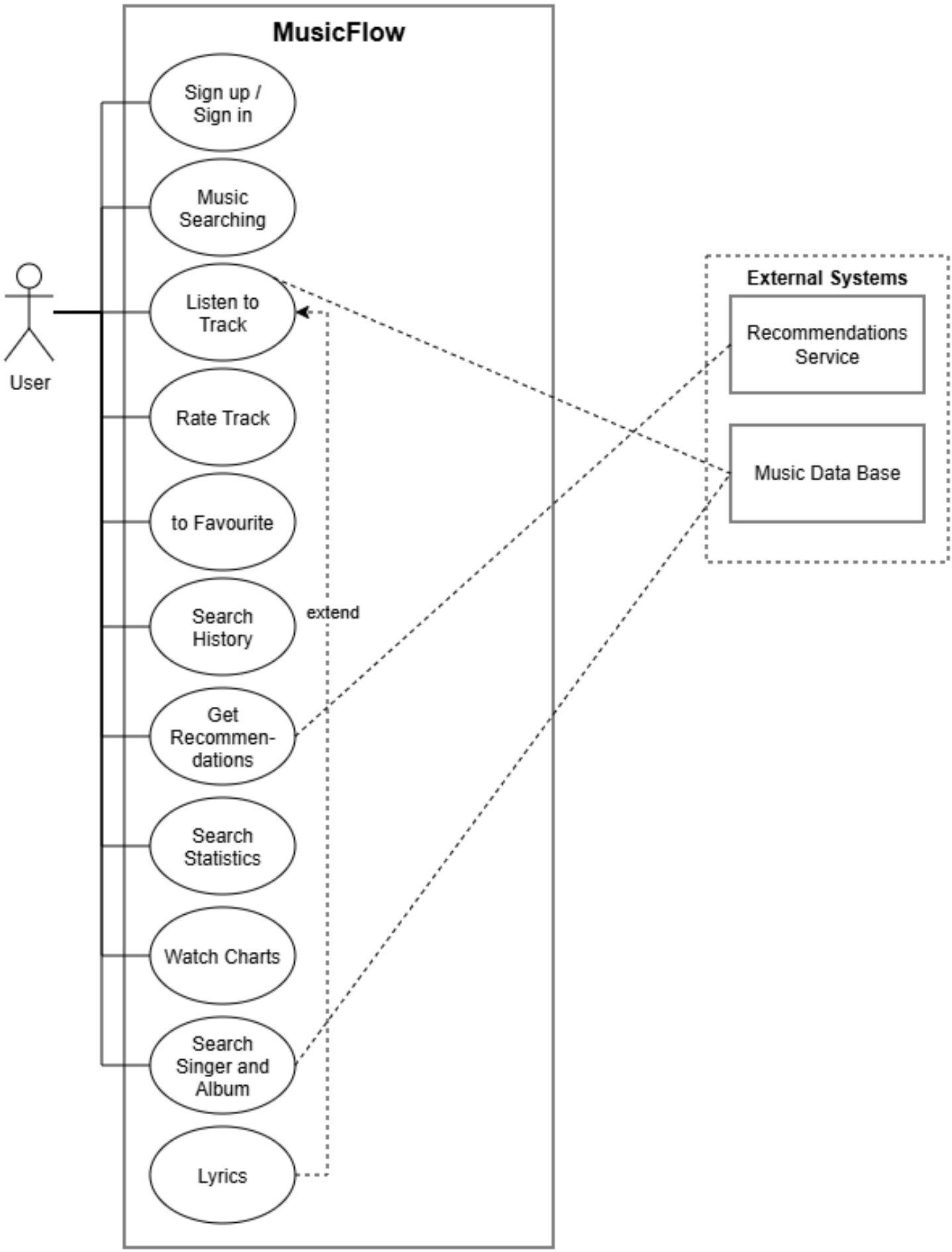
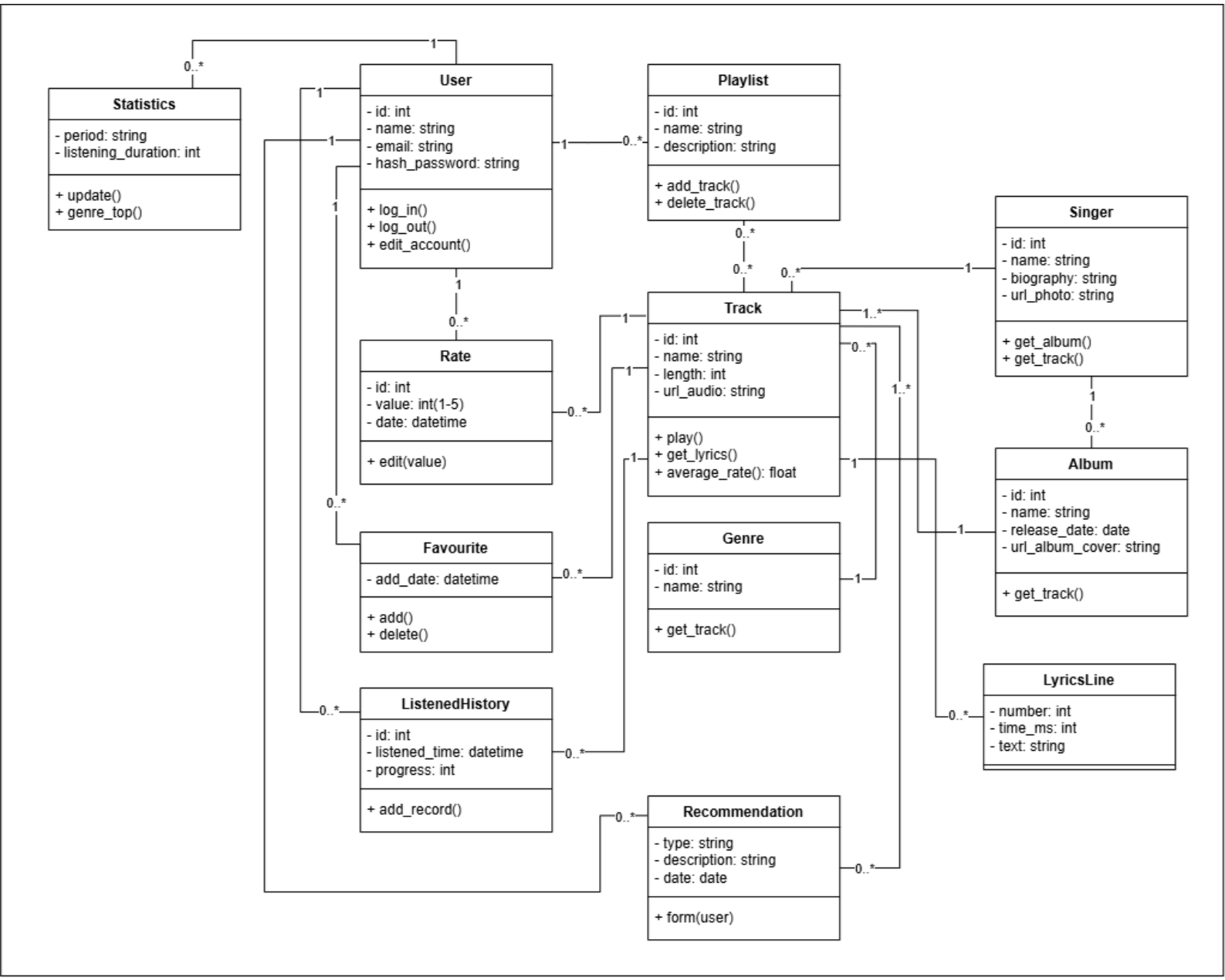
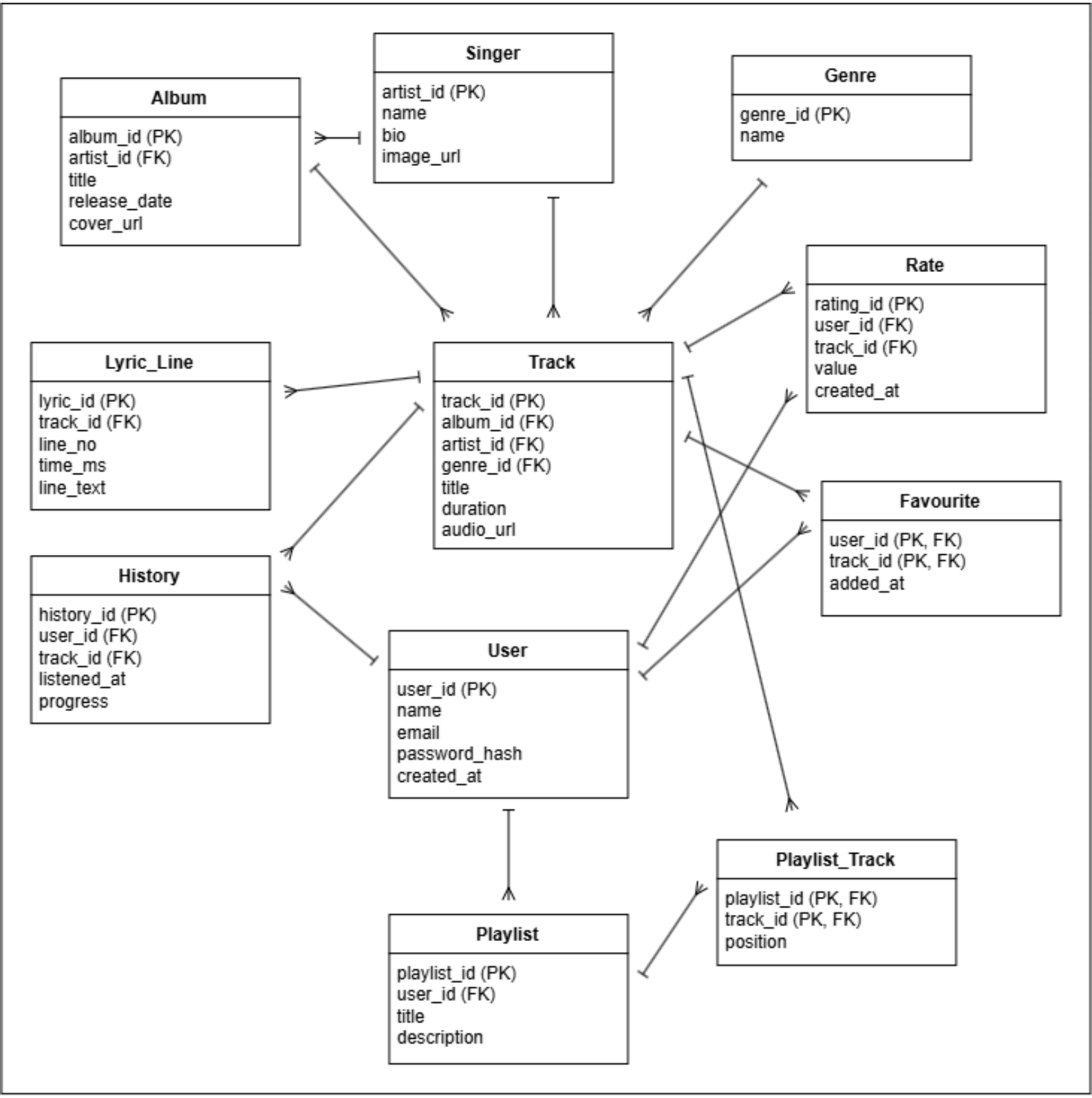
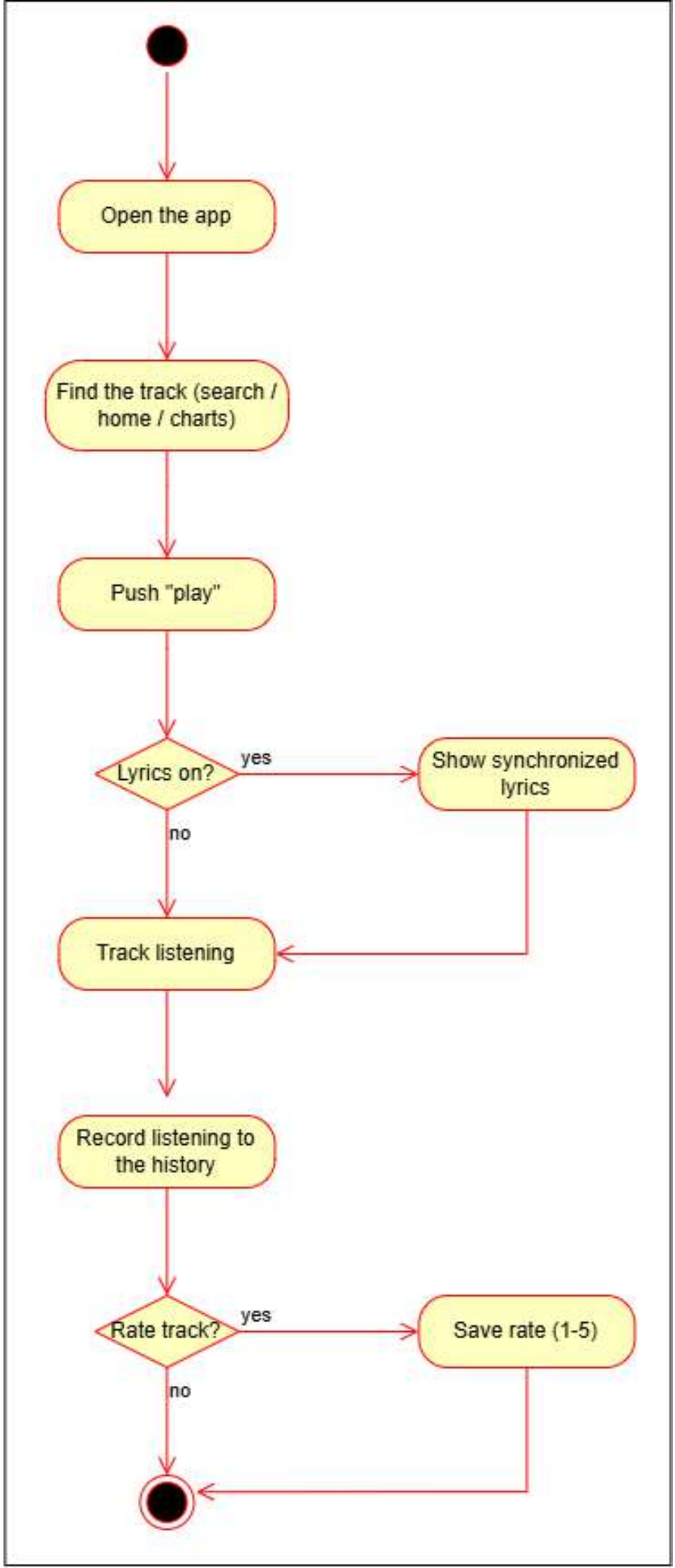
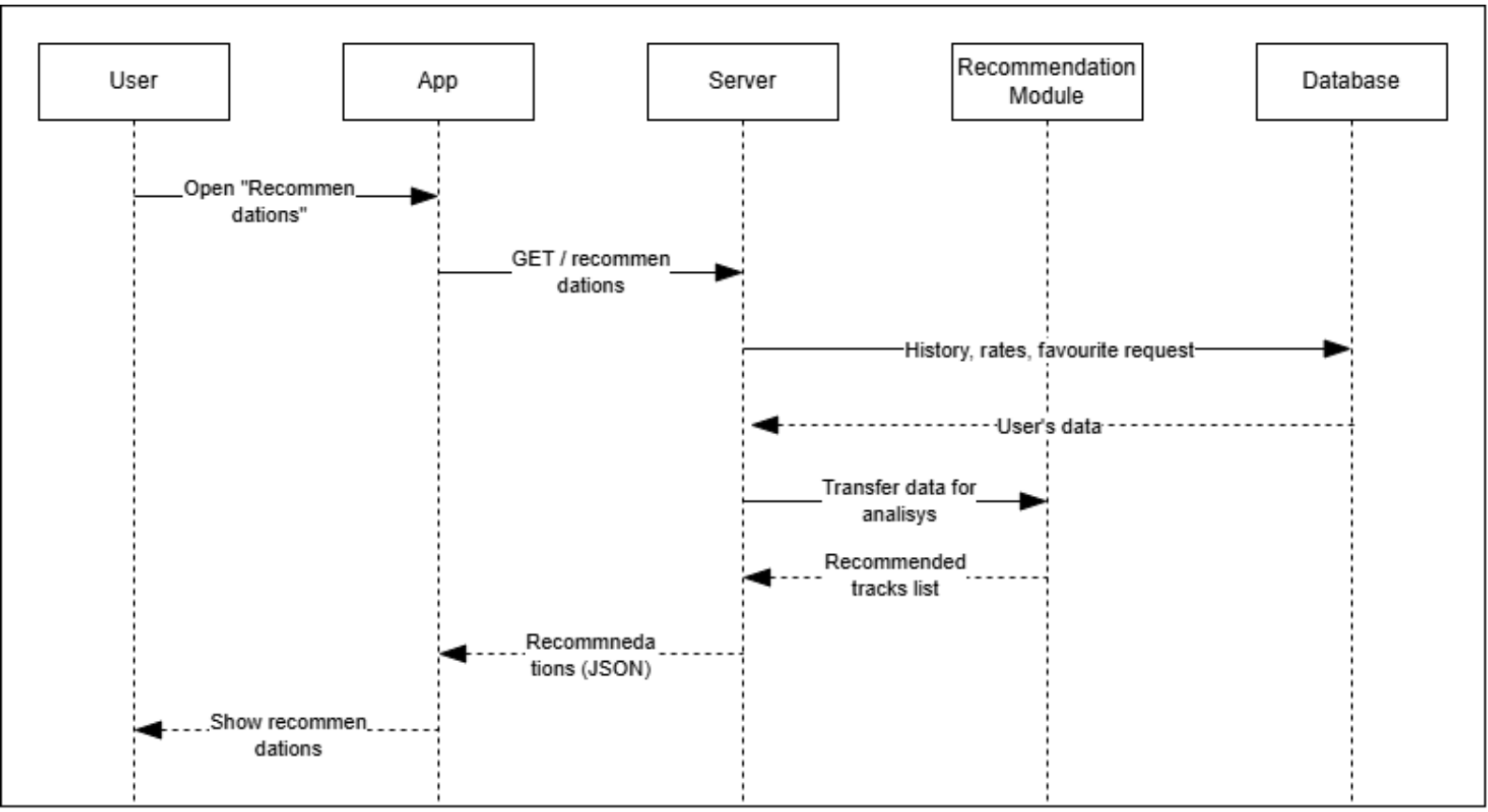

Client App
(Flutter)
Screens:
Home
Search
Player
Text
Charts
Statistics
Charts
Client REST API
AudioPlayer and Lyrics synchronization
HTTPS / JSON
Server Part
(REST API)
Autorization module
Catalogue (tracks, albums, singer,search)
Playlists, Favourite, Rates
History, Statistics
Charts, Recommen-dations
Database
(PostgreSQL)
External music API / audio storage


MusicFlow
Sign up / Sign in
Music Searching
Listen to Track
Rate Track
to Favourite
Search History
Get Recommen-dations
Search Statistics
Watch Charts
Search Singer and Album
Lyrics
User
extend
External Systems
Recommendations Service
Music Data Base


Statistics
- period: string
- listening_duration: int
+ update()
+ genre_top()
User
- id: int
- name: string
- email: string
- hash_password: string
+ log_in()
+ log_out()
+ edit_account()
Playlist
- id: int
- name: string
- description: string
+ add_track()
+ delete_track()
Singer
- id: int
- name: string
- biography: string
- url_photo: string
+ get_album()
+ get_track()
Track
- id: int
- name: string
- length: int
- url_audio: string
+ play()
+ get_lyrics()
+ average_rate(): float
Rate
- id: int
- value: int(1-5)
- date: datetime
+ edit(value)
Album
- id: int
- name: string
- release_date: date
- url_album_cover: string
+ get_track()
Favourite
- add_date: datetime
+ add()
+ delete()
Genre
- id: int
- name: string
+ get_track()
LyricsLine
- number: int
- time_ms: int
- text: string
ListenedHistory
- id: int
- listened_time: datetime
- progress: int
+ add_record()
Recommendation
- type: string
- description: string
- date: date
+ form(user)
1
0..*
1..*


Singer
artist_id (PK)
name
bio
image_url
Genre
genre_id (PK)
name
Album
album_id (PK)
artist_id (FK)
title
release_date
cover_url
Rate
rating_id (PK)
user_id (FK)
track_id (FK)
value
created_at
Lyric_Line
lyric_id (PK)
track_id (FK)
line_no
time_ms
line_text
Track
track_id (PK)
album_id (FK)
artist_id (FK)
genre_id (FK)
title
duration
audio_url
Favourite
user_id (PK, FK)
track_id (PK, FK)
added_at
History
history_id (PK)
user_id (FK)
track_id (FK)
listened_at
progress
User
user_id (PK)
name
email
password_hash
created_at
Playlist_Track
playlist_id (PK, FK)
track_id (PK, FK)
position
Playlist
playlist_id (PK)
user_id (FK)
title
description


Open the app
Find the track (search / home / charts)
Push "play"
Lyrics on?
yes
Show synchronized lyrics
no
Track listening
Record listening to the history
Rate track?
yes
Save rate (1-5)
no


User
App
Server
Recommendation Module
Database
Open "Recommen dations"
GET / recommen dations
History, rates, favourite request
User's data
Transfer data for analisys
Recommended tracks list
Recommneda tions (JSON)
Show recommen dations